Propositional Logic

Recap:

- Propositions $\rightarrow$ atomic propositions / compound propositions.
- Logical connectors

  $\wedge, \vee, \sim(\neg), \oplus, \Rightarrow, \Leftrightarrow$
- Truth tables
- Tautology

Given two propositions p, q

connector	compound proposition	
$\wedge$	$p \wedge q$	True, if p and q are true.
$\vee$	$p \vee q$	True, If p or q is true.
$\neg$	$\neg p$	true, If p is false.

- Any proposition can be denoted using a boolean variable

  p: sky is blue, $\neg p$: sky is not blue.

Syntax vs Semantic in propositional logic

Syntax → whether given Proposition is gramatically correct

Semantic → meaning of a gramatically correct statement.

$1 \subseteq \mathbb{Z}$ gramatically incorrect ✗

$1 \in \mathbb{Z}$ gramaticaly correct — 1 is an Integer True.

$2 \notin \mathbb{Z}$ gramaticaly correct — false

Def Implication ($\Rightarrow$) (This is a logical connector)

Given two propositions p and q, we can create compound proposition $p \Rightarrow q$ (denoted as 'p implies q'), is true if the truth of p implies truth of q.

p: It rains in Bozeman today

q: Bozeman is wet today.

$p \Rightarrow q$

If it rains in Bozeman today, then Bozeman is wet today.

In order for this compound proposition to be true, Bozeman must be wet today whenever it rains in Bozeman today.

$p \Rightarrow q$

The p is called antecedent (or premise / hypothesis)

The q is called consequent (or the conclusion)

$p: 1+1=2$ (T) $q: 2+3=6$ (F)

$r = (p \Rightarrow q)$

If $1+1=2$ then $2+3=6$

What can you say about truth value of $p \Rightarrow q$ False

$s: 1+1=3$ (F) $t: 2+2=4$ (T)

$s \Rightarrow t$ True

$a: 1+1=2$ $b: 1+3=4$

$a \Rightarrow b$ True

$c: 1+1=6$ $d: 2+3=10$

$c \Rightarrow d$ True.

Remember! In an implication premise and conclusion do not need to be related in order for it to be true. What matters is whenever the premise is true, the conclusion is true.

$$p \Rightarrow q \equiv \neg p \vee q$$

If $2+2=5$, then moon is made out of cheese

Boy

Girl.

Babies wear diapers

I am not a baby.

Affirming consequent fallacy.

$p \Rightarrow q$ (is true)

$q \Rightarrow p$ (this is not correct)

Def Exclusive or ($\oplus$)

Given two propositions p and q, the compound proposition $p \oplus q$ ("p exclusive or q", p xor q) is true, when one of p and q is true but not both.

In other words, $p \oplus q$ is false, whenever p and q is true or p and q is false.

p: Alex is holding a cup of tea on his right hand.

q: Alex is holding a cup of coffee on his right hand.

Consider $r : p \oplus q$

r: Alex is holding a cup of tea on his right hand or a cup of coffee on his right hand, but not both.

Note: In English, it is very hard to distinguish between "inclusive or" and "exclusive or"

Def If and only if ($\Leftrightarrow$, Iff)

Given two propositions p and q, the compound proposition $p \Leftrightarrow q$ ("p if and only if q) is true when propositions p and q have the same truth value.

$$p \Leftrightarrow q \equiv (p \Rightarrow q) \wedge (q \Rightarrow p)$$

$n \in \mathbb{Z}$

p: The integer n is divisible by 4

q: The integer n^2 is divisibly by 16.

Claim: $p \Leftrightarrow q$

n is divisible by 4 iff n^2 is divisible by 16.

If I properly set my alarm, then I will get up at correct time.

I will wake up at correct time if and only if I properly set my alarm,

Truth Tables

Def

A truth table (TT) for a given proposition lists for each possible truth assignment for that proposition (with one truth assignment per row), the truth value of the entire proposition

ex: $(p \vee r) \wedge (p \vee q)$

\# variables is 3

\# unique truth assignments for 3 variables

$= 2 \times 2 \times 2 = 2^3$

$= 8$ //

If a proposition has n variables, then the # of unique truth assignments is 2^n

Ex: Given two propositions p, q, let's consider several compound propositions

p	q	$p \vee q$	$p \wedge q$	$p \oplus q$	$p \Rightarrow q$	$p \Leftrightarrow q$	$\neg(p \oplus q)$	$\neg p$	$\neg p \vee q$
0	0	0	0	0	1	1	1	1	1
0	1	1	0	1	1	0	0	1	1
1	0	1	0	1	0	0	0	0	0
1	1	1	1	0	1	1	1	0	1

$$p \Leftrightarrow q \equiv \neg(p \oplus q)$$

$$p \Rightarrow q \equiv \neg p \vee q$$

$q \Rightarrow p$	$(p \Rightarrow q) \wedge (q \Rightarrow p)$
1	1
0	0
1	0
1	1

Def Tautology

A proposition is a tautology if it is true under every possible truth assignment.

Ex: $p \Rightarrow p$

\# variables 1

\# unique truth assignments $= 2^1 = 2$

p	$p \Rightarrow p$
0	1
1	1

This is a tautology.

Is $(p \wedge (p \Rightarrow q)) \Rightarrow q$ is a tautology

variables 2

unique Truth assignments $4 = 2^2$

p	q	$p \Rightarrow q$	$p \wedge (p \Rightarrow q)$	$(p \wedge (p \Rightarrow q)) \Rightarrow q$
0	0	1	0	1
0	1	1	0	1
1	0	0	0	1
1	1	1	1	1

This is a tautology.

Is $(p \vee q) \wedge (p \vee r) \wedge (p \oplus q)$ a tautology?

\# variables = 3

\# unique TA = $2^3 = 8$

p	q	r	$p \vee q$	$p \vee r$	$p \oplus q$	$(p \vee q) \wedge (p \vee r) \wedge (p \oplus q)$
0	0	0	0	0	0	0
0	0	1	0	1	0	0
0	1	0	1	0	1	0
0	1	1	1	1	1	1
1	0	0	1	1	1	1
1	0	1	1	1	1	1
1	1	0	1	1	0	0
1	1	1	1	1	0	0

This is not a tautology.